

Reusable compression for the complex finite network

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Abstract

We give two reusable methods for compressing weighted side computations while retaining the binary phase-frame contract in OpenAI’s integer multiplication manuscript. Coordinate enclosures support disjoint-triple rectangles; an explicit orthonormal family supports shared intersection-two sums. Signed transparent computation restores arbitrary scratch, and an even-ground-size matching reuses complete first/third-stage auxiliary banks. At $h = 26$ the construction supports complex saving $5 \cdot 10^{-9}$, exceeding the retained bit saving. An independent combined upstream patch integrates this construction and the retained compact-control arguments, supporting the conditional bound $\kappa = 2^{-31}$. The previously published 2^{-34} artifacts are unchanged.

The retained manuscript is pinned at

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

Its algorithmic interfaces remain assumptions. This local research note concerns the same fixed finite-alphabet, fixed-tape model and makes no practical runtime or unrestricted optimality claim.

1 Weighted side computations with binary phase frames

We retain the complex motif’s data banks, central operations, three tensor stages, and endpoint corrections. Scalars lie in $\mathbb{Z}[i, 1/2]$ and labels in $F = \mathbb{F}_2^h$ with its standard dot product. Throughout this construction $h \geq 8$ is even. Write $\mathcal{T} = \binom{[h]}{3}$, $v = \binom{h}{3}$, $N = v^3$ and $m = h^3$.

For triples S, T , the central operation has coefficient $(|S \cap T| - 1)/2$. Its side correction has coefficient $+1/2$ for disjoint triples and $-1/2$ for intersection-two triples. Intersection-one pairs need no side correction; the diagonal central coefficient is one. We compress these two side relations separately. All coefficients below are characteristic-zero coefficients; binary labels do not turn the scalar arithmetic into XOR arithmetic.

1.1 A reusable weighted circuit transfer

An all-ones rectangle with a sources and b targets uses $a + b - 1$ roles. Allocate a input slots and b output slots sharing one pivot. Add the other inputs into the pivot, then add it into the other outputs. This invertible mixer L has an all-ones output-by-input submatrix. Its inverse undoes these additions in reverse order. A fixed dyadic rectangle weight is placed in the output injection J , not in the binary labels.

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More generally, a circuit of disjoint-support binary sums with c addition nodes and q output uses has an invertible embedding on $c + q$ roles. Allocate one outgoing role per use of each input. At an addition, reuse its first incoming role as one outgoing pivot, allocate the other outgoing roles freshly, add the second input to the pivot, and fan it out. The second incoming role retires. There are $2c + q$ uses and c role identifications, proving the count. Starting from zero side values, disjoint input supports give the exact integer coefficients, not merely their reductions modulo two. Equal sums may have shared fanout.

Let V copy data into the input slots, let J inject weighted outputs, and retain the central gather G and scatter R . The signed chronological schedule is

$$L, -J, L^{-1}, -R, V, G, R, L, J, L^{-1}, -G, -V.$$

An arbitrary side vector z contributes $-JLz + JL(z + Vx) = JLVx$. The two inverse mixers and final negative copy restore z . Similarly the centers contribute RGx and return to their original values. Thus the invocation is $y \leftarrow y + x$ precisely when $JLV + RG = I$. Reversing and inverting the signed operations implements $y \leftarrow y - x$.

For the frame transfer use the original decomposition $A = B \perp P$, where P is the earlier tensor-indicator line, and suppress the future norm-one tensor line Q . Put

$$D_0 = B \otimes F, \quad D_1 = A \otimes F, \quad D_U = (B \otimes F) \perp (P \otimes U),$$

$$X_t = D_{\langle t \rangle}, \quad Y_t = D_{t^\perp}.$$

The forward schedule's common gate frames, in the displayed order, are

$$D_0, D_0, D_0, D_0, X_t, D_1, D_0, D_U, Y_t, D_1, D_1, D_1.$$

Copy and injection gates are grouped by data triple. In a rectangle the middle mixer has one common U . In a sum circuit it has one U per node. For the inverse invocation with logical banks exchanged, the frames are

$$D_0, D_0, D_0, X_t, D_U, D_1, D_0, Y_t, D_1, D_1, D_1, D_1.$$

The inverse middle mixer uses physical- X target supports reachable from each forward node. These grow along reversed edges; forward source supports must not be substituted for them.

A valid assignment must make every side-role transition increasing and nondegenerate, with each nonzero orthogonal residual nonalternating. Merely having nested source sets is insufficient over $\mathbb{F}_2$. The following two constructions explicitly meet this stronger condition in both directions.

1.2 Disjoint triples: coordinate enclosures

Partition the ordered disjoint-triple relation into rectangles $\mathcal{S} \times \mathcal{U}$. Their ground-point unions are disjoint. In a forward middle mixer choose its frame U as follows:

- If $\mathcal{S} = \{S\}$, use $U = \langle t_S \rangle$.
- Otherwise, if $\mathcal{U} = \{T\}$, use $U = t_T^\perp$.
- Otherwise, use the coordinate subspace on $\bigcup \mathcal{S}$.

For the inverse invocation interchange the two families in this rule. In every case $\langle t_S \rangle \subset U \subset t_T^\perp$ for every rectangle pair. These spaces are nondegenerate. In the two singleton cases the only new nontrivial local complement is that of two orthogonal triple lines. It has a norm-one coordinate vector outside their six-point union, since $h > 6$.

In the third case both unions have at least four points. The complement of any source triple line within U contains a coordinate unit outside that triple. The complement of U within $t_T^\perp$ contains a coordinate unit in the target union outside T . The spaces U and $U^\perp$ are coordinate spaces. Consequently all nonzero residuals in $0 \subset \langle t_S \rangle \subset U \subset t_T^\perp \subset F$, and all skipped transitions actually used by input-only or output-only roles, have norm-one vectors. Every such nondegenerate residual has an orthonormal basis by the retained binary-space lemma.

The partition is explicit. For disjoint a - and b -subsets of $[n]$, $0 \leq a, b \leq 3$, allow row stars, column stars, and a split into l and $n - l$ points. Split cases by the source and target cardinalities in the first part, and use the product of the already chosen smaller partitions. For a plan let (C, L, R) be its rectangle count and source/target membership counts. Products multiply these counts coordinatewise and cases add them. An empty relation has zero counts; if one subset size is zero, use the full rectangle. Try all split sizes and choose the least $L + R - C$, breaking ties by row stars, then column stars, then increasing split size. This finite recursion is a construction, not a global rectangle optimality claim.

At $h = 26$, the counts are

$$(C, L, R) = (24870, 217255, 299571), \quad R_0 = L + R - C = 491956.$$

The exact checker verifies that each of the 4604600 ordered disjoint triple pairs appears once. It also checks the frame-case hypotheses for every generated rectangle. Every such rectangle has scalar weight $+1/2$.

1.3 Intersection two: an orthonormal sum compiler

Fix the unique common pair $P = \{i, j\}$. The source and target triples are $P \cup \{a\}$ and $P \cup \{b\}$ with $a \neq b$ outside P . The binary vectors

$$t_a = e_i + e_j + e_a \quad (a \notin P)$$

are pairwise orthonormal. Since h is even, adjoining

$$r_i = e_i + \sum_{a \notin P} e_a, \quad r_j = e_j + \sum_{a \notin P} e_a$$

gives an orthonormal basis of the entire F : each extra vector has odd weight $h - 1$, their mutual product is $h - 2 = 0$ in $\mathbb{F}_2$, and both are orthogonal to every t_a .

Compute each leave-one-out sum with prefix and suffix sums on the $n = h - 2$ inputs. Omit unused totals and intern equal input-support sets. The active circuit has $3n - 6$ additions and n outputs, hence $4n - 6$ roles. Each sum is cancellation-free over the integers. Inject each output with weight $-1/2$. Distinct triples intersecting in two points have a unique common pair, so these circuits give every required coefficient exactly once.

At a forward node let U span its ancestor vectors t_a ; in the inverse middle circuit let U span the reachable output vectors t_b . Nested support differences are spanned by subsets of an orthonormal family. No source can reach its excluded output. At the final transition into the relevant $t_b^\perp$, the residual has an orthonormal basis consisting of the remaining t_a and the two extra vectors

r_i, r_j . Cleanup complements have the same property. Thus every required residual is zero or nonalternating and nondegenerate in both directions.

This argument applies to any disjoint-support circuit for the same relation, not only prefix/suffix sums. It is a reusable phase-frame transfer criterion. At $h = 26$ the role count is

$$R_2 = \binom{26}{2}(4 \cdot 24 - 6) = 29250, \quad R_{\text{side}} = R_0 + R_2 = 521206.$$

2 Whole-network composition and accounting

All local frames above are tensored with the original future line Q . The original data boundaries remain $A \otimes \langle t \rangle \otimes Q$ to $D_1 \otimes Q$ on X and $B \otimes \langle t \rangle \otimes Q$ to $Y_t \otimes Q$ on Y . Their residuals and the original tensor complements have norm-one witnesses as in the retained proof. Tensoring nonalternating nondegenerate factors preserves a norm-one witness. All new side transitions are increasing. Only the $h+1$ centers decrease once per invocation, by dimension h . Thus $L = 3v^2h(h+1)$ is unchanged.

Pair consecutive ground points and put $\pi(T) = \{x \oplus 1 : x \in T\}$. This is an involution of triples. A triple with a full ground-point pair intersects its image in two points; otherwise the intersection is empty. Hence $t_T \perp t_{\pi(T)}$. Match the complete stage-1 auxiliary bank at fixed triples (A, B) to the stage-3 bank at $(B, \pi(A))$, preserving each local role index. These banks have equal size, the matching is bijective, and no simultaneous roles are identified. Each invocation restores arbitrary scratch, so its scalar correctness is preserved.

All auxiliary first frames are $D_0 \otimes Q$ and all last frames are $D_1 \otimes Q$. The joining labels are therefore

$$E = F \otimes \langle t_A \rangle \otimes \langle t_B \rangle, \quad H = \langle t_B \otimes t_{\pi(A)} \rangle^\perp \otimes F.$$

They are nondegenerate and $E \subset H$. The residual $H \cap E^\perp$ contains a norm-one coordinate tensor whose second coordinate lies outside $A \cup \pi(A)$; such a coordinate exists because $h > 6$. Thus the join also has an orthonormal residual basis. Replacing the old edges $E \rightarrow F^{\otimes 3}$ and $0 \rightarrow H$ by $E \rightarrow H$ saves exactly m residual dimensions per removed role and introduces no decrease.

Consequently the entire network has

$$W = 2N + 2v^2(R_{\text{side}} + h + 1), \quad s = Wm - 2N + 2L.$$

Forward, inverse, forward stages send (X, Y) to $(-Y, X)$ and restore every surviving auxiliary role. Data endpoint labels and all scratch endpoints $0, F^{\otimes 3}$ are unchanged. The retained weight-27 phase corrections, signed-bank correction and return permutation therefore give the same $C^{\otimes m}$ operator on every role, for any number of columns. Each orthonormal residual contributes precisely its dimension in forward or inverse translation kernels. These are the full complex-interface obligations.

At $h = 26$,

$$m = 17576, \quad v = 2600, \quad N = 17576000000, \quad L = 14236560000,$$

$$W = 7082222160000, \quad s = 124477130005280000, \quad Wm - s = 6678880000, \quad \eta = \frac{19}{354111108}.$$

Exact logarithm bounds give $\eta > (5/10^9) \log m$, and hence $s/W < m^{1-5/10^9}$. Thus this construction supports $a_c = 1 - \sigma = 5 \cdot 10^{-9}$. The exact saving is approximately $5.48945 \cdot 10^{-9}$. The finite even- h screen is not an all- h ceiling.

3 Conditional assembly and precision

Keep the published bit saving $a_b = 296/10^{11}$ and the supplied compact-control movement proof. Choose

$$\epsilon = 199/1000, \quad c = 1, \quad \beta = 1/100, \quad \zeta = 1/1000, \quad \delta = 1/10000, \quad C_1 = 4961/1000,$$

$$\lambda = 1 - 293/10^{11}, \quad \lambda' = 1 - 29/10^{10}, \quad \kappa = 2^{-31}.$$

Here $\sigma < \tau$, so the internal exponent is $\chi = \tau$. The leaf saving is $(1 - \beta)a_c > a_b$ and the reservation exponent is zero. The seven retained margins have exact minimum

$$G_* = \epsilon(1 - \lambda') = 5771/10^{13} > 2^{-31}.$$

All parameter inequalities have strict slack. The compact reserved-axis cost is now $O(V \log p)$, and $\epsilon(1 + c) < 1$ still holds.

The new scalar mixers use only additions, subtractions and half scalings. For explicit guard accounting let $M_0 = R_0 - C$ be the number of additions in all disjoint-rectangle mixers per invocation. Each pair circuit has $2(3n - 6)$ mixer additions. Let M be their combined mixer count, V_0 the number of input-copy incidences, and J_0 the output incidences. The signed schedule uses at most

$$T_{\text{scalar}} = 3v^2(4M + 2V_0 + 4J_0 + 18v)$$

elementary scalar operations, including half scalings. The exact checker verifies $4T_{\text{scalar}} + 8s + 8W + 8m < 64(W + m + 1)^3$ and $2 \leq s < m^5$. Thus the retained additive node charge $E = 64(W + m + 1)^3$ dominates these operations, per-kernel scalar corrections, and fixed endpoint corrections. Binary basis changes only permute coefficient encodings. The generalized stopped-depth proof applies with the new $W, m, s, C_1 = 5 - 4\beta + \zeta$ and its explicit C_0 . Since $\epsilon C_1 < 1$, coefficient widths remain $O(p)$.

This conditional construction is integrated in the independent combined patch listed below, including the new complex constants, generalized guard and final assembly. The previously published 2^{-34} note and patch remain unchanged. Neither the computations nor this note provide independent review or a formal verification of the upstream theorem or the new transfer.

4 Artifacts

- `scripts/complex_compression.py`: construction and exact accounting.
- `scripts/binary_phase_frames.py`: reusable binary-space checker.
- `patches/complex-compression-31.patch`: complete independent patch against the pinned original, generated by `scripts/make_complex_compression_patch.py`.
- `certificates/complex-compression.json`: construction-source hash, full disjoint-pair enumeration, finite frame controls, matching, network counts and conditional assembly margins.
- `tests/test_complex_compression.py`: exact rational scalar maps with independent dirty variables, both signed directions, shared scratch composition, phase identities, and rejection of alternating or degenerate residuals.
- `tests/test_complex_compression_patch.py`: source dependencies, separate arities, exact parameter substitution and reference checks.

Finite examples do not replace the general proofs.